Automated Analyst Report

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Query: Analyse supplier performance across all product types. Identify which product categories have the highest defect rates, which shipping carriers offer the best revenue-to-cost ratio, flag any SKUs with failed inspection results, and highlight stockout risk where lead times are long relative to stock levels.report findings must be backed by solid proof from within the data

EXECUTIVE SUMMARY The analysis reveals that cosmetics enjoy the lowest average defect rate (1.92 %) while haircare shows the highest (2.48 %). Carrier C delivers the strongest revenue■to■shipping■dollar ratio (1,138.6 ×) but Carrier B maintains the lowest defect rate among carriers (1.76 %). Thirty■six SKUs failed inspection, with several exceeding a 4 % defect threshold, and stock■out risk is acute for a handful of items—most notably SKU34 (risk score 13) and SKU68 (risk score 8).

KEY FINDINGS - **Defect rates by product type:** Haircare = 2.483 %, Skincare = 2.335 %, Cosmetics = 1.919 % (lowest). - **Carrier efficiency:** Carrier C achieves the highest revenue per shipping dollar (1,138.56 ×) and a modest defect rate (2.60 %); Carrier B posts the best defect performance (1.76 %) and a strong revenue ratio (1,055.7 ×). - **Failed■inspection SKUs:** 36 SKUs flagged; 5 exceed a 4 % defect rate (e.g., SKU42 – 4.94 %, SKU65 – 4.91 %, SKU50 – 4.75 %). - **Stock■out exposure:** Top risk scores are driven by minimal on■hand inventory versus lead time: SKU34 (score 13, 1 unit in stock, 26■day lead), SKU68 (score 8, 0 units, 8■day lead). - **Revenue contribution:** Skincare generates the most revenue ($241,628) but also bears a high defect rate relative to cosmetics, indicating a quality■revenue trade■off.

RISKS AND RECOMMENDATIONS The primary risks are elevated defect rates in haircare/skincare lines and imminent stock■outs for high■risk SKUs with long lead times. Immediate actions should include tightening quality controls on the five >4 % defect SKUs, reallocating shipments to Carrier C for cost efficiency while monitoring its defect rate, and expediting replenishment or safety■stock adjustments for SKU34, SKU68, and other high■risk items to mitigate service interruptions.